

# An AI-Powered Inclusive Mobile Messaging System for Users with Visual, Hearing, and Speech Disabilities

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**Abstract**—The digital exclusion of people with sensory disabilities from mobile messaging platforms represents one of the most pressing challenges in human-centred software engineering. This article presents the design, implementation, and empirical evaluation of an AI-powered inclusive mobile messaging system specifically conceived for users with visual and hearing disabilities. The proposed architecture integrates an intelligent processing API that orchestrates text-to-speech (TTS), speech-to-text (STT), automatic image description via computer vision, and optical character recognition (OCR) modules, dynamically adapting each message format to the receiver’s sensory profile. The technology stack combines React Native for a cross-platform mobile interface, Spring Boot (Java) for business logic, Django (Python) for AI processing, RabbitMQ as a message broker, and PostgreSQL for persistence, with end-to-end asymmetric-key encryption. Development followed the TDDM4IoTS methodology (*Test-Driven Development Methodology for IoT-Based Systems*). The system was evaluated with  $n=25$  participants (60 % with visual impairment, 20 % with hearing impairment, and 20 % without disability) applying the Technology Acceptance Model (TAM) and a WCAG 2.1 checklist. Results revealed a global TAM mean of  $\bar{x}=4.20/5$  ( $\bar{x}_{PU}=4.55$ ;  $\bar{x}_{PEOU}=3.85$ ), strong internal consistency ( $\alpha=0.866$ ; PU:  $\alpha=0.867$ ; PEOU:  $\alpha=0.898$ ), and the highest-rated item reaching 84 % maximum-scale responses. An iterative accessibility improvement cycle produced a 64 % reduction in WCAG 2.1 warnings. The findings validate that strategic AI integration in mobile applications constitutes a viable approach to bridging the communicative gap of people with sensory disabilities.

**Index Terms**—Accessibility, assistive technology, end-to-end encryption, human-machine interaction, image captioning, inclusive messaging, mobile application, React Native, sensory disability, speech recognition, speech synthesis, TAM, TDDM4IoTS, WCAG.

## I. INTRODUCTION

**I**NSTANTANEOUS communication has become a fundamental right in contemporary society and a cornerstone of full civic participation. Despite mobile messaging platforms surpassing three billion active users globally, their proliferation has not been matched by intrinsic accessibility architectures that guarantee equitable use. According to the World Health

Organization (WHO), approximately 15 % of the world population lives with some form of disability, a significant proportion of which involves visual, hearing, or speech impairments [1]. In Ecuador, the National Council for Disability Equality (CONADIS) registers 124,801 individuals with active sensory disabilities: hearing impairments account for 50.30 %, visual impairments for 45.39 %, and speech impairments for 4.31 %, with a gender distribution of 70,429 men (56.4 %) and 54,372 women (43.6 %), according to CONADIS institutional records [2].

The most widely used messaging platforms—WhatsApp, Telegram, Messenger, and Instagram—offer only partial accessibility features: limited screen-reader compatibility (TalkBack and VoiceOver), contrast adjustment, and, in the case of WhatsApp, recently introduced audio-to-text transcription. However, none of these platforms comprehensively addresses the needs of users with multiple sensory profiles: bidirectional adaptation among text, audio, and image modalities; automatic semantic description of visual content; complete navigation by voice commands; nor end-to-end encryption for accessible multimodal data flows [3], [4].

From a Human-Machine Interaction (HMI) perspective, universal design [5] and the Web Content Accessibility Guidelines (WCAG 2.1) [6] require that systems be perceivable, operable, understandable, and robust for the widest possible range of users. In the mobile context, systematic accessibility evaluations of Android and iOS applications reveal non-compliance rates exceeding 60 % in criteria such as text alternatives for non-text content and complete navigation without precise touch interaction [7], [8].

Recent advances in Artificial Intelligence (AI) offer unprecedented opportunities to overcome these barriers. State-of-the-art deep neural network TTS systems produce naturalistic speech synthesis [9], while transformer-based speech-to-text models have achieved word error rates (WER) comparable to human transcription [10]. Simultaneously, image captioning models have demonstrated practical utility for users with visual disabilities in real-world scenarios [11].

Against this background, the present work is motivated by the following research questions:

- RQ1** Can a single mobile platform coherently integrate TTS, STT, image captioning, and OCR modules—together with asymmetric end-to-end encryption—while maintaining zero processing error rates under variable input conditions?

**RQ2** Does the proposed system achieve a Technology Acceptance Model score above the positive acceptance threshold of  $\bar{x} > 4.0$  in a heterogeneous population of individuals with sensory disabilities?

**RQ3** Can the TDDM4IoTS development methodology, applied with accessibility as a formal acceptance criterion, produce a measurable and statistically meaningful reduction in WCAG 2.1 warnings compared to leading commercial messaging platforms?

The contributions of this work are:

- An accessibility-oriented microservices architecture that integrates TTS, STT, OCR, image description, and asymmetric end-to-end encryption in a cross-platform mobile messaging system (addressing RQ1).
- A systematic application of the TDDM4IoTS methodology to inclusive messaging software development [12].
- Empirical evidence on technological acceptance—measured via TAM—and technical accessibility from a study involving 25 participants with heterogeneous sensory disabilities (addressing RQ2).
- Quantification of the impact of an iterative accessibility improvement process, achieving a global 64 % reduction in warnings identified by Google’s Accessibility Scanner (addressing RQ3).

The remainder of this article is organized as follows. Section II reviews related work in three dimensions: accessibility in mobile messaging, augmentative and alternative communication (AAC), and AI for accessibility, identifying the gaps that motivate this research. Section III describes the mixed research paradigm, the eleven TDDM4IoTS phases applied—including the microservices architecture, intelligent message conversion flow, voice command module, asymmetric encryption, and hybrid infrastructure—and the 18 WCAG 2.1 accessibility criteria employed. Section IV presents technical API performance results (TTS, STT, image description, video), iterative accessibility evaluation results, and TAM questionnaire results. Section V interprets the findings in the context of existing literature and reflects on SDG implications. Section VI synthesises the main conclusions and directions for future research.

## II. RELATED WORK

### A. Accessibility in Mobile Messaging Applications

1) *Empirical evaluations with real users*: Weichbroth [4] conducted a systematic review of 57 usability studies on mobile applications, concluding that accessibility is among the least addressed criteria in conventional design processes. Chen *et al.* [7] carried out a large-scale empirical study of Android app accessibility—including popular messaging platforms—revealing non-compliance rates exceeding 60 % in fundamental WCAG 2.1 criteria such as text alternatives for non-text content, semantic labelling, and screen-reader compatibility.

Da Silva *et al.* [13] performed the first systematic evaluation of WhatsApp from the perspective of visually impaired users, identifying 18 critical barriers including absent alternative texts, insufficient auditory feedback on notifications, and low

visibility of interactive control states. Zahida *et al.* [14] extended this approach to elderly users, documenting that absent persistent font-size adjustment, icon density on small screens, and insufficient haptic feedback are primary error sources. Swati *et al.* [15] found that Facebook Messenger is systematically inaccessible for users with complete blindness due to missing accessibility labels on critical buttons. Rumini and Astrid [16] identified that aesthetic and consistency dimensions present the greatest violations in Telegram according to Nielsen’s heuristics. Kim *et al.* [17] found significant deficiencies in the emotional and affective interface design of mobile communication apps for hearing-impaired users.

2) *Systematic reviews and accessibility frameworks*: Balantyne *et al.* [18] compared W3C, Google, Apple, and other normative accessibility guidelines, concluding that cross-platform fragmentation creates inconsistencies and recommending WCAG 2.1 as the transversal reference. Acosta-Vargas *et al.* [8] demonstrated that a combined automated-plus-manual approach increases detection coverage by more than 35 %. Huang *et al.* [19] evaluated the four most widely used messaging platforms with adults with mild-to-moderate cognitive disabilities, documenting systematic failures in touch-target size, colour contrast, and screen orientation. Bercaru and Popescu [3] conducted a PRISMA review of 2,819 articles concluding that adaptive personalisation of user experience remains the most deficient dimension in accessibility research.

### B. Augmentative and Alternative Communication Systems

1) *Conceptual framework and base technologies*: Elsharhar *et al.* [20] systematised modalities of AAC high-technology platforms and concluded that machine learning represents the most promising avenue, particularly through speech generation, text prediction, and biological signal classification. Guerrero-Ulloa *et al.* [21] presented BellChat, the first inclusive web messaging application integrating simultaneous TTS and STT for users with visual and hearing disabilities, though limited to a web-only environment. Yeratziotis *et al.* [22] integrated sign language into social media apps for deaf users; however, their approach does not address visual disability or speech impairment. Chang *et al.* [23] proposed a smart city system combining gesture and voice recognition for hearing- and visually-impaired citizens. Janaka *et al.* [24] explored ubiquitous messaging via smart glasses, demonstrating that voice input reduces response time by 33.1 % and increases typing speed by 40.3 %.

2) *AI-based AAC applications for speech disorders*: Costanzo *et al.* [25] evaluated Talkitt—an ML application that translates unintelligible sounds into clear words—with children with Down syndrome, showing recognition precision improved from 75 % to 90 % after personalised calibration. Mateos-Sanchez *et al.* [26] developed CapacitaBOT, an IBM Watson conversational chatbot for people with intellectual disabilities. Bleakley *et al.* [27] documented critical barriers in commercial STT systems for people who stammer: excessively short end-of-utterance detection windows and inability to manage disfluencies, with direct implications for inclusive STT module design.

### C. Artificial Intelligence for Accessibility

Chemnad and Othman [28] conducted a PRISMA bibliometric review of 3,706 articles (2018–2023), presenting a classification framework for AI in accessibility and concluding that research is concentrated asymmetrically on visual disabilities while hearing, speech, cognitive, and motor disabilities receive significantly less coverage. Ji *et al.* [29] and Tan *et al.* [9] established the state of the art in neural TTS, while Baevski *et al.* [10] and Radford *et al.* [30] demonstrated WER comparable to human transcription in STT systems. Ming *et al.* [31] published the most comprehensive review of automatic image captioning to date. Dognin *et al.* [11] demonstrated on the VizWiz dataset—composed of images taken by blind people in real-world conditions—that modern captioning systems can reach utility levels comparable to human descriptions in accessibility contexts. Mukherjee *et al.* [32] showed OCR accuracy exceeding 95% for standard printed characters. Saleem *et al.* [33], Papatsimouli *et al.* [34], and Kothadiya *et al.* [35] advanced real-time sign language recognition systems.

### D. Identified Gaps

The critical analysis of the 35 reviewed works identifies five structural gaps that justify the contribution of this work.

**Gap 1—Absence of multimodal mobile platforms simultaneously covering visual, hearing, and speech profiles.** No reviewed work implements a mobile messaging system that dynamically adapts each message to the receiver’s sensory profile combining TTS, STT, image captioning, and OCR in a single native mobile platform.

**Gap 2—Technological fragmentation:** mature AI components exist in isolation but have not been integrated in a coherent system with guaranteed privacy. No work combines TTS, STT, captioning, and OCR in a messaging architecture with asymmetric end-to-end encryption.

**Gap 3—Scarcity of empirical TAM evidence** with heterogeneous multi-disability populations. Existing studies are restricted to single disability profiles.

**Gap 4—Absence of systematic, test-driven development methodologies** for accessible messaging software. TDDM4IoTS has not previously been adapted to inclusive mobile messaging development.

**Gap 5—Insufficient coverage of speech disorder profiles** in voice-input messaging systems. No existing system offers automatic input-mode adaptation according to the user’s individual disfluency profile.

This work directly addresses all five identified gaps.

## III. MATERIALS AND METHODS

This section describes, first, the materials employed—platforms, instruments, and AI components—and, second, the methods applied for system development and for technical and user evaluation. The study adopted a mixed-methods research paradigm [36]: quantitative for API conversion performance metrics (mean latency, error rate) and for the TAM questionnaire; qualitative for the analysis of accessibility barriers identified during direct user observation sessions. The research

Table I  
TECHNOLOGY STACK OF THE INCLUSIVE MESSAGING SYSTEM

| Component      | Technology           | Justification                                     |
|----------------|----------------------|---------------------------------------------------|
| Mobile UI      | React Native         | Cross-platform; native TalkBack/VoiceOver support |
| REST backend   | Spring Boot          | Microservices, ORM, horizontal scalability        |
| AI API         | Django (Python)      | ML/NLP ecosystem, pre-trained models              |
| Real-time msg. | WebSocket + RabbitMQ | Ordered delivery, fault tolerance                 |
| Persistence    | PostgreSQL           | Relational, ACID-compliant, open source           |
| Containers     | Docker + Nginx       | Portability, isolation, reverse proxy             |
| Cloud (AI)     | AWS Windows Server   | CPU/RAM for model inference                       |
| Encryption     | Asymmetric keys      | End-to-end confidentiality                        |

type is applied and projective [36], oriented toward the design, implementation, and evaluation of a software artifact that solves a specific technical and social problem in inclusive communication.

### A. Materials

1) *Hardware Infrastructure and Deployment:* The system was deployed on a hybrid infrastructure consisting of two complementary nodes. The **AI node** is an Amazon Web Services (AWS) instance running Windows Server, assigned to the AI API (Django/Python) given its greater CPU and RAM demands for vision and language model inference. The **backend node** is a CentOS server administered by UTEQ, hosting the REST API (Spring Boot/Java), the RabbitMQ message broker, and the PostgreSQL database. Nginx served as a web server and reverse proxy on both nodes; Docker containerization ensured portability and service isolation between development and production environments. Technical API tests were executed from an Android mobile device connected to a standard WiFi network (802.11n, −55 dBm).

2) *Technology Stack:* Table I summarises the software components, selected technology, and technical justification for each choice. React Native was selected for the mobile interface for its ability to generate native UI components for Android and iOS from a single codebase, reducing maintenance costs and ensuring compatibility with platform accessibility services (TalkBack on Android, VoiceOver on iOS).

3) *Artificial Intelligence Components:* The accessible conversion modules integrate the following AI components. For **TTS**, two engines were integrated in parallel: Edge-TTS (Microsoft Neural TTS, multilingual, cloud-based) and SAPI5 (Microsoft Speech API 5, local processing via pyttsx3). For **STT**, a deep-learning transformer-based automatic speech recognition API [10], [30] was used (Whisper, executed on the AWS node). For **image description**, pre-trained vision-language models were integrated via API calls, specifically blip-image-captioning-base from Hugging Face, consistent with the state of the art [11], [37]. For **OCR**, computer vision models with accuracy exceeding 95% on standard printed text [32] were used. For **voice command**

**navigation**, cosine similarity [38] was used as a semantic proximity metric, with cascade correction via the Llama 1B language model and a phonetic dictionary.

4) *Accessibility Evaluation Framework*: The accessibility evaluation framework was built from three complementary normative sources: WCAG 2.1 [6] at conformance levels A and AA, W3C mobile accessibility guidelines, and Android/iOS guidelines systematised by Ballantyne *et al.* [18] and Acosta-Vargas *et al.* [8]. From these sources, 18 verifiable criteria were defined, organised according to the WCAG POUR principles: (i) **Perceivability**: transcription of audio and video messages, text alternatives for images, configurable synthesised speech rate, and minimum colour contrast of 4.5:1 for normal text (WCAG criterion 1.4.3); (ii) **Operability**: minimum touch target size of 48×48 dp, adaptable screen orientation, complete voice command navigation; (iii) **Understandability**: informative error suggestions, explicitly declared interface language, icon and label consistency; (iv) **Robustness**: verified compatibility with TalkBack and VoiceOver, and semantic marking of all interactive controls. The automated evaluation tool was Google’s Accessibility Scanner, applied to the seven main system screens.

5) *Technological Acceptance Instrument*: Technological acceptance was measured using the TAM proposed by Davis [39], in its classic two-construct formulation, widely validated in accessible mobile application studies [40]. The questionnaire comprised 12 items (PTAM001–PTAM012) on a 5-point Likert scale (1=strongly disagree; 5=strongly agree), organised into two constructs: **Perceived Usefulness (PU)**—items PTAM001–PTAM006—and **Perceived Ease of Use (PEOU)**—items PTAM007–PTAM012. The questionnaire was administered in Google Forms with TTS support for visually impaired participants. Internal reliability was verified using Cronbach’s alpha, calculated with R Studio (packages *psych*, *ggplot2*, and *BiplotR*, version 4.3.1), yielding  $\alpha=0.866$  globally,  $\alpha=0.867$  for PU, and  $\alpha=0.898$  for PEOU—all above the 0.80 threshold indicative of good internal consistency [41].

## B. Methods

1) *Development Framework: TDDM4IoTS*: The system was developed using TDDM4IoTS [12], which integrates test-driven development (TDD), model-driven engineering (MDE), and Scrum/XP agile principles into eleven iterative phases designed for distributed systems combining heterogeneous hardware and software components. Its selection over conventional methodologies responds to three reasons: (1) Phase 5 (*Test Generation*) incorporates accessibility as a verifiable criterion before code generation, surpassing the reactive model of conventional methodologies where accessibility tests are post-hoc activities; (2) its iterative nature allows incorporation of user-test findings without architectural restructuring; (3) its tool-agnostic design permits selection of the most appropriate technology for each deliverable [12]. Table II describes how the eleven phases were applied to the inclusive messaging system.

Table II  
APPLICATION OF TDDM4IoTS TO THE INCLUSIVE MESSAGING SYSTEM:  
PHASES, DELIVERABLES, AND DESIGN DECISIONS

| P. | Phase                   | Deliverable                                                        | Design decision                               |
|----|-------------------------|--------------------------------------------------------------------|-----------------------------------------------|
| 1  | Preliminary analysis    | Checklist (18 WCAG 2.1 criteria); tech. analysis                   | Edge-TTS + SAPI5; AWS + UTEQ infrastructure   |
| 2  | Technology layer design | 3-tier microservices architecture                                  | Separate REST / AI API nodes                  |
| 3  | Requirements analysis   | 5 use cases with UML diagrams                                      | 5 main flows + user profiles                  |
| 4  | Models                  | Class, state, activity, DB diagrams                                | PostgreSQL relational model                   |
| 5  | Test generation         | Unit/integration tests; TAM survey; accessibility protocol         | Acceptance criterion: error rate = 0          |
| 6  | Software generation     | REST API (Spring Boot), AI API (Django), mobile app (React Native) | WebSocket + RabbitMQ                          |
| 7  | Model refinement        | Multimedia architecture update                                     | HTTP <i>multipart</i> for messages >64 KB     |
| 8  | Software refinement     | Refactored code ( <i>clean code</i> )                              | Removal of duplicate IDs in virtualised lists |
| 9  | Deployment              | Production system                                                  | Docker + Nginx on two nodes                   |
| 10 | Deliverable evaluation  | Requirements compliance report                                     | Final validation of 18 WCAG 2.1 criteria      |
| 11 | Feedback                | Incident and improvement log                                       | Correction of 48 of 75 accessibility warnings |

2) *System Design: Architecture and Modules*: The system adopts a three-tier decoupled microservices architecture, illustrated in Fig. 1. The **presentation layer** is the React Native mobile interface. The **business layer** comprises two independent services: the Spring Boot REST API, managing authentication, message routing, and encryption; and the Django AI API, concentrating the accessible conversion models. The **persistence layer** is the PostgreSQL database. Real-time communication is managed via WebSocket with RabbitMQ as a message broker. For multimedia messages exceeding the WebSocket 64 KB frame limit, HTTP with *multipart/form-data* was adopted as an alternative protocol (Phase 7 of TDDM4IoTS).

The intelligent conversion module operates in three steps, as illustrated in Fig. 2: (1) the Spring Boot backend receives the message, decrypts it with the sender’s private key, and forwards it to the Django API according to MIME type (text, audio, image, or video); (2) the Django API applies the transformation to the receiver’s declared sensory profile: TTS if the receiver has a visual impairment, STT if hearing impaired, vision-language model description plus OCR if the message is an image and the receiver has a visual impairment; (3) the processed message is re-encrypted with the receiver’s key and delivered via WebSocket. End-to-end encryption uses asymmetric private keys: each user receives a unique key stored locally on their mobile device, so the server never stores plain text.

The voice command navigation module allows complete operation of the mobile interface via spoken instructions. The STT-transcribed audio is compared against the registered com-





Table III  
COMPARATIVE PERFORMANCE OF TTS ENGINES EDGE-TTS AND SAPI5 BY TEXT LENGTH. MEAN OF THREE REPETITIONS; ERROR RATE = 0 IN ALL CASES.

| Edge-TTS (Microsoft Neural TTS – cloud) |       |        |       |
|-----------------------------------------|-------|--------|-------|
| Metric                                  | Short | Medium | Long  |
| Load time (s)                           | 8.13  | 7.90   | 16.88 |
| Net latency (s)                         | 8.10  | 7.90   | 16.87 |
| Send size (KB)                          | 0.37  | 0.54   | 1.03  |
| Response size (MB)                      | 0.16  | 0.30   | 0.75  |
| Errors                                  | 0     | 0      | 0     |

| SAPI5 (pyttsx3 – local) |       |        |       |
|-------------------------|-------|--------|-------|
| Metric                  | Short | Medium | Long  |
| Load time (s)           | 5.18  | 5.76   | 15.07 |
| Net latency (s)         | 5.14  | 5.73   | 15.05 |
| Send size (KB)          | 0.37  | 0.54   | 1.03  |
| Response size (MB)      | 1.46  | 2.93   | 7.24  |
| Errors                  | 0     | 0      | 0     |

autonomously, assisted by TTS in case of visual impairment.

c) *Statistical analysis.*: Internal consistency was verified using Cronbach’s alpha. For each item, the arithmetic mean ( $\bar{x}$ ), standard deviation (SD), and Likert-level frequency distribution were calculated. The factorial structure of the instrument was explored via an HJ-Biplot graph [43], [44], which allowed identification of item clusters with high internal correlation and participants with similar response patterns. All analyses were performed in R Studio (v. 4.3.1; packages psych, ggplot2, and BiplotR) and Microsoft Excel.

#### IV. EVALUATION AND RESULTS

##### A. Technical Performance of the Conversion API

The conversion API was subjected to controlled load testing to characterise latency, stability, and processing integrity in each module. All test conditions were executed from an Android mobile device on a standard WiFi network (802.11n, –55 dBm), with three repetitions per condition, reporting the arithmetic mean. The technical acceptance criterion defined in TDDM4IoTS Phase 5 was: error rate equal to zero for all input conditions.

1) *Text-to-Speech (TTS) Module*: The TTS module integrates two parallel engines: Edge-TTS (Microsoft Neural TTS, cloud-based) and SAPI5 (local, via pyttsx3). Three text length categories were evaluated: short (151 chars), medium (317 chars), and long (799 chars). Table III presents the complete results.

Both engines operated with zero error rates across all conditions, verifying the acceptance criterion. SAPI5 exhibited lower latencies than Edge-TTS at all text lengths: 5.14 s vs. 8.10 s for short text (37 % lower), 5.73 s vs. 7.90 s for medium text (27 % lower), and 15.05 s vs. 16.87 s for long text (11 % lower), explained by the absence of network latency in SAPI5’s local synthesis. However, SAPI5 generates substantially larger response files: 7.24 MB for long text vs. 0.75 MB for Edge-TTS, which may compromise efficiency in bandwidth-limited mobile environments. The minimum

Table IV  
PERFORMANCE OF THE STT MODULE (WHISPER) BY AUDIO DURATION. ERROR RATE = 0 IN ALL CASES.

| Metric              | Short  | Medium | Long   |
|---------------------|--------|--------|--------|
| Load time (s)       | 13.361 | 20.512 | 55.000 |
| Connection time (s) | 0.003  | 0.008  | 0.011  |
| Net latency (s)     | 13.361 | 20.509 | 54.990 |
| Send (KB)           | 60.382 | 233.50 | 1000   |
| Response (KB)       | 60.723 | 233.85 | 1000   |
| Errors              | 0      | 0      | 0      |

Table V  
PERFORMANCE OF THE BLIP IMAGE DESCRIPTION MODULE BY RESOLUTION. ERROR RATE = 0. LAT. = NET LATENCY (S); SEND = SEND SIZE (MB); RESP. = RESPONSE SIZE (MB).

| Resolution (px) | Weight  | Lat.  | Send | Resp. |
|-----------------|---------|-------|------|-------|
| 640×427         | 44 KB   | 13.95 | 0.06 | 0.12  |
| 1280×853        | 126 KB  | 17.67 | 0.17 | 0.24  |
| 1920×1280       | 254 KB  | 16.42 | 0.35 | 0.41  |
| 6000×4000       | 1.65 MB | 19.19 | 2.31 | 2.37  |

observed latency (5.14 s for SAPI5, short text) exceeds the 2 s threshold that the literature identifies as optimal for natural conversational flow [9], representing the most critical technical limitation of the system in its current state.

2) *Automatic Speech Recognition (STT) Module*: The STT module uses the Whisper model [30], executed on the AWS node. Three audio files representative of messaging user behaviour were evaluated [45]. Table IV presents the detailed results.

Latency grows approximately linearly with audio duration: 13.36 s for short audio, 20.51 s for medium (53 % higher), and 54.99 s for long (311 % higher relative to short). Send and response sizes are practically identical across all conditions (difference <1 %), confirming processing integrity.

3) *Image Description Module (Image Captioning)*: The image description module uses the blip-image-captioning-base model from Hugging Face. Four resolutions representative of current mobile cameras were evaluated, from 640×427 px (44 KB) to 6,000×4,000 px (1.65 MB). Table V presents the results.

No errors were recorded for any evaluated resolution. Latency ranged from 13.95 s to 19.19 s (37 % relative variation across the full resolution range). Notably, the 1,920×1,280 px image exhibited lower latency (16.42 s) than the 1,280×853 px image (17.67 s), suggesting that inference time variability is driven by image semantic complexity rather than resolution per se [11], [37].

4) *Video Processing Module*: The video module manages large multimedia file transmission via HTTP *multipart/form-data*, adopted in TDDM4IoTS Phase 7 to overcome WebSocket’s 64 KB frame limit. Three representative video durations were evaluated: short (4 min, 9.22 MB), medium (17 min, 48.9 MB), and long (30 min, 76.9 MB), all at 360p. Table VI presents the complete results.

Load times scaled linearly with duration: 3.91 min, 20.29 min, and 34.00 min. A noteworthy finding is that effec-

Table VI

PERFORMANCE OF THE VIDEO MODULE BY DURATION. EFFECTIVE TRANSMISSION SIZE INCLUDES BASE64 ENCODING OVERHEAD ( $\approx 40\%$  OVER ORIGINAL FILE WEIGHT). ERROR RATE = 0 IN ALL CASES.

| Metric              | Short (4min)   | Medium (17 min) | Long (30 min)  |
|---------------------|----------------|-----------------|----------------|
| Resolution          | 360p           | 360p            | 360p           |
| Original file (MB)  | 9.22           | 48.9            | 76.9           |
| Load time (min)     | 3.91           | 20.29           | 34.00          |
| Connection time (s) | 0.15           | 1.35            | 1.354          |
| Net latency (min)   | 3.91           | 20.29           | 34.00          |
| Send – base64 (MB)  | 12.89          | 68.45           | 107.55         |
| Response (MB)       | 12.91          | 68.50           | 107.68         |
| Send–response diff. | 0.155 %        | 0.077 %         | 0.121 %        |
| Base64 overhead     | $\approx 40\%$ | $\approx 40\%$  | $\approx 40\%$ |
| Errors              | 0              | 0               | 0              |

Table VII

REDUCTION IN ACCESSIBILITY WARNINGS PER SCREEN FOLLOWING THE TDDM4IoTS ITERATIVE IMPROVEMENT CYCLE (PHASES 5 AND 11).

| Screen       | Initial   | Final     | Reduction | %           |
|--------------|-----------|-----------|-----------|-------------|
| Main         | 11        | 2         | 9         | 81.8        |
| User profile | 21        | 8         | 13        | 61.9        |
| Settings     | 9         | 2         | 7         | 77.8        |
| Messaging    | 21        | 13        | 8         | 38.1        |
| Camera       | 4         | 0         | 4         | 100.0       |
| Photo        | 2         | 0         | 2         | 100.0       |
| Video        | 7         | 2         | 5         | 71.4        |
| <b>Total</b> | <b>75</b> | <b>27</b> | <b>48</b> | <b>64.0</b> |

tive transmission sizes (12.89 MB, 68.45 MB, and 107.55 MB) exceeded original file weights (9.22 MB, 48.9 MB, and 76.9 MB) by approximately 40 %, attributable to base64 encoding required by HTTP *multipart/form-data*. This overhead should be considered when dimensioning bandwidth and managing user progress notifications. The send-to-response difference was below 0.2 % in all cases, confirming transmission integrity.

### B. Accessibility Evaluation

The accessibility evaluation was conducted in two phases using Google’s Accessibility Scanner [8] on the seven main system screens, in line with the combined automated evaluation protocol proposed by Acosta-Vargas *et al.* [8]. Table VII presents the comparison.

The iterative cycle eliminated 48 of the 75 initial warnings, achieving a global reduction of 64.0 %. Resolved warnings primarily involved: (i) touch elements below the minimum  $48 \times 48$  dp (operability criterion); (ii) colour contrast below 4.5:1 (WCAG 1.4.3); (iii) text not exposed in accessibility labels (robustness criterion); (iv) absent or ambiguous element descriptions. Camera and photo screens achieved total warning elimination (100 %), while settings and main screens reached 77.8 % and 81.8 % reductions, respectively. The messaging screen recorded the lowest reduction (38.1 %), for an architectural reason: reuse of virtualised list templates in React Native

Table VIII

TAM QUESTIONNAIRE RESULTS ( $n=25$ , LIKERT 1–5 SCALE). PU = PERCEIVED USEFULNESS (PTAM001–006); PEOU = PERCEIVED EASE OF USE (PTAM007–012). CONSTRUCT MEAN COMPUTED AS ARITHMETIC AVERAGE OF ITEM MEANS. SD COMPUTED WITH SAMPLE ESTIMATOR ( $N - 1$ ).  $\alpha$ : CRONBACH’S ALPHA (R PSYCH PACKAGE, v. 4.3.1); 95 % CI VIA BOOTSTRAPPING (2,000 ITERATIONS).

| Item      | Construct | $\bar{x}$ | SD                            | Lv. 5 (%)                     |
|-----------|-----------|-----------|-------------------------------|-------------------------------|
| PTAM001   | PU        | 4.48      | 1.005                         | 72                            |
| PTAM002   | PU        | 4.76      | 0.597                         | 84                            |
| PTAM003   | PU        | 4.48      | 0.770                         | 64                            |
| PTAM004   | PU        | 4.52      | 0.770                         | 64                            |
| PTAM005   | PU        | 4.56      | 0.768                         | 68                            |
| PTAM006   | PU        | 4.52      | 0.872                         | 72                            |
| PU mean   |           | 4.55      | $\alpha=0.867$ [0.579, 0.957] |                               |
| PTAM007   | PEOU      | 3.56      | 1.357                         | 40                            |
| PTAM008   | PEOU      | 3.96      | 1.207                         | 48                            |
| PTAM009   | PEOU      | 4.20      | 1.000                         | 52                            |
| PTAM010   | PEOU      | 4.16      | 1.068                         | 52                            |
| PTAM011   | PEOU      | 3.72      | 1.208                         | 40                            |
| PTAM012   | PEOU      | 3.48      | 1.229                         | 32                            |
| PEOU mean |           | 3.85      | $\alpha=0.898$ [0.803, 0.948] |                               |
| Global    |           | 4.20      | 0.95                          | $\alpha=0.866$ [0.693, 0.932] |

generates duplicate accessibility identifiers that cannot be resolved without refactoring towards dynamic unique identifiers per list element. This result compares favourably with the  $>60\%$  non-compliance levels documented by Chen *et al.* [7] in the most widely used commercial messaging platforms.

### C. Technological Acceptance: TAM Results

Technological acceptance was quantified using the 12-item TAM questionnaire administered to the  $n=25$  participants. Table VIII presents means ( $\bar{x}$ ) and standard deviations (SD) per item. Internal consistency was strong:  $\alpha=0.866$  globally (95 % CI: [0.693, 0.932]),  $\alpha=0.867$  ([0.579, 0.957]) for PU, and  $\alpha=0.898$  ([0.803, 0.948]) for PEOU—all exceeding the 0.80 threshold for good internal consistency [41]. Confidence intervals were obtained by bootstrapping (2,000 iterations).

1) *Perceived Usefulness*: The PU construct (PTAM001–PTAM006) achieved the highest construct mean ( $\bar{x}_{PU}=4.55$ ). Item PTAM002 (“Using this application would improve my effectiveness in communicating with others”) reached the highest rating ( $\bar{x}=4.76$ ; SD=0.60), with 84 % of participants at the maximum level, indicating that AI-mediated communicative effectiveness is the most practically valued contribution. Item PTAM001 ( $\bar{x}=4.48$ ; SD=1.005; 72 % at level 5) reflects that the system enables faster daily communication. Items PTAM003–PTAM006 present means between 4.48 and 4.56 with low variability ( $SD \leq 0.87$ ), indicating that communicative productivity, message exchange effectiveness, interaction facilitation, and general utility are consistently and positively valued across all evaluated profiles.

2) *Perceived Ease of Use*: The PEOU construct (PTAM007–PTAM012) achieved a construct mean of  $\bar{x}_{PEOU}=3.85$ , with notably greater internal dispersion than PU. Items PTAM009 ( $\bar{x}=4.20$ ;  $SD=1.00$ ) and PTAM010 ( $\bar{x}=4.16$ ;  $SD=1.07$ )—evaluating clarity and interaction flexibility—show positive ratings indicating that the system’s main interface is perceived as comprehensible. Items PTAM007 ( $\bar{x}=3.56$ ;  $SD=1.36$ ) and PTAM012 ( $\bar{x}=3.48$ ;  $SD=1.23$ ) obtained the two lowest scores. PTAM007 (“Learning to use the application was easy for me”) concentrates the greatest questionnaire variability (full range 1–5), indicating that the initial learning curve was heterogeneous across profiles—consistent with Bleakley *et al.* [27], who document steeper learning curves for voice-input paradigms among users with low technology experience (40 % of the sample). The domain of all system functions requires a longer appropriation process than the 20-minute evaluation protocol allowed.

3) *Multivariate Analysis: HJ-Biplot*: The bidimensional HJ-Biplot representation retains 67.8 % of the total variance of the TAM response matrix (Dimension 1: 43.6 %; Dimension 2: 24.2 %), exceeding the 60 % threshold recommended in the literature for a 2D projection to be considered sufficiently faithful to the original multivariate structure [43], [44].

The HJ-Biplot graph (Fig. 3) revealed two structural patterns of interest. First, the vectors of items PTAM001–PTAM006 (PU) and PTAM008–PTAM010 (partial PEOU) point in a similar direction in the left quadrant, indicating high internal correlation and that participants with greater affinity to this cluster perceived both utility and interactional clarity coherently. Second, the vectors of PTAM007 and PTAM012 are approximately orthogonal to the main cluster, revealing that these items capture independent variance associated with initial learning ease and complete functional mastery, respectively. Participants 8, 10, 17, 18, and 25 are positioned in the right quadrant, representing lower affinity with all TAM constructs; this subgroup coincided in direct observation with participants recording the greatest number of erroneous taps. The global instrument mean was  $\bar{x}=4.20$  ( $SD=0.95$ ), exceeding the positive acceptance threshold of  $\bar{x}>4.0$  established in TAM studies in mobile accessibility contexts [40]. This value was obtained as the arithmetic mean of the 300 individual records ( $n=25 \times 12$  items), equivalent by the balanced instrument design (6 items per construct) to the mean of the twelve item means and to the mean of the two construct means.

4) *Direct Observation*: The direct observation session recorded the following behavioural indicators: 7 interaction errors (taps on incorrect elements), 2 requests for facilitator assistance, and 15 total erroneous screen taps. All three indicators were concentrated primarily on the messaging and camera screens during tasks T2 and T4, consistent with the higher density of interactive elements and the residual accessibility warnings documented in Table VII. The low number of help requests (2 of 25 participants, 8 %) confirms that the majority of users completed the predefined tasks autonomously.

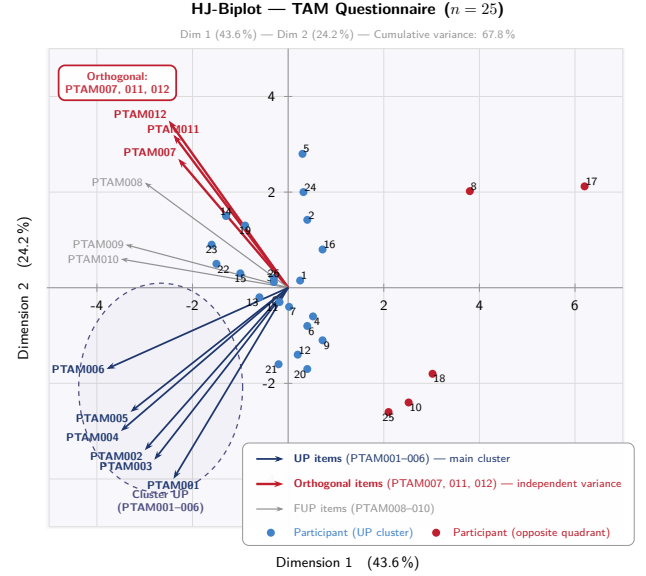


Figure 3. HJ-Biplot of TAM questionnaire responses ( $n=25$ ). Dimension 1 explains 43.6 % of variance; Dimension 2 explains 24.2 % (cumulative: 67.8 %). Arrows indicate item vectors; proximity of participant points to a vector reflects agreement with that item. The PU cluster (PTAM001–006) and most PEOU items (PTAM008–010) point in the same direction, indicating high internal correlation. PTAM007 and PTAM012 are orthogonal to the main cluster, capturing independent variance.

Table IX  
RESULTS SUMMARY BY EVALUATION DIMENSION AND GAP ADDRESSED.

| Dimension       | Main result                                                                                                               | Gap addressed |
|-----------------|---------------------------------------------------------------------------------------------------------------------------|---------------|
| API performance | Error rate = 0 across TTS, STT, captioning, and video; min. TTS latency: 5.14 s; base64 overhead in video: $\approx 40$ % | Gaps 1 and 2  |
| Accessibility   | 64 % reduction in WCAG 2.1 warnings (48/75); 100 % on camera and photo                                                    | Gap 4         |
| TAM acceptance  | $\bar{x}_{global}=4.20$ ; PU = 4.55; PEOU = 3.85; $\alpha=0.866$                                                          | Gap 3         |
| Observation     | 8 % assistance requests; errors concentrated on messaging                                                                 | Gaps 1 and 5  |

## D. Results Summary

Table IX provides an integrated view of the results obtained across the three evaluated dimensions, linking each result to the identified gap it addresses.

## V. DISCUSSION

### A. Interpretation of Technical Results

The four conversion modules—TTS, STT, image description, and video processing—operated with zero error rates

across all evaluated conditions, validating the stability of the proposed microservices architecture and satisfying the TDDM4IoTS Phase 5 acceptance criterion (RQ1 = confirmed). This result is relevant because none of the inclusive messaging systems reviewed in the literature—BellChat [21], Stefanov *et al.* [46], nor GlassMessaging [24]—reports API robustness evaluations with variable input conditions.

The minimum observed TTS latency was 5.14 s (SAPI5, short text), exceeding the 2 s threshold that Tan *et al.* [9] identify as optimal for natural conversational flow. This limitation is not unique to the proposed system: Bercaru and Popescu [3] document that inference latency remains the most critical deployment barrier for neural TTS models in real production environments, and that only implementations with predictive caching or non-autoregressive models achieve latencies below 2 s. The advantage of SAPI5 over Edge-TTS in net latency (37 % lower for short text) is explained by the absence of network latency in local synthesis; this advantage reverses in bandwidth-constrained mobile environments, where SAPI5 response files (7.24 MB for long text vs. 0.75 MB for Edge-TTS) represent a significant transfer penalty. This latency-versus-response-volume trade-off had not previously been documented in the context of inclusive mobile messaging systems and constitutes an operationally relevant finding for future implementations.

STT latency scaled from 13.36 s for short audio to 54.99 s for long audio, with perfect transmission integrity (< 1 % difference between send and response sizes). This behaviour is consistent with Whisper’s architecture [30], in which the global attention mechanism over the complete audio sequence determines that computational cost is proportional to duration. Latency reduction via dedicated GPU acceleration or distilled models (Whisper Tiny/Small) constitutes the highest-impact practical improvement identified.

The non-monotonic latency variability in the image description module—1,920×1,280 px recorded lower latency than 1,280×853 px—is consistent with Dognin *et al.* [11], who document that semantic difficulty rather than resolution is the primary inference time predictor. This implies that latency prediction in an inclusive messaging application cannot rely solely on image size metadata, requiring a semantic complexity estimator for managing user expectations.

A noteworthy operational finding is the approximately 40 % base64 encoding overhead in video transmission. This increment must be considered in bandwidth dimensioning and in user progress notifications during file transfer. Transmission integrity was confirmed in all cases (< 0.2 % send-to-response difference).

### B. Interpretation of Accessibility Results

The global 64.0 % reduction in accessibility warnings (75 to 27) after the TDDM4IoTS iterative cycle places the proposed system notably above the commercial platform baseline (RQ3 = confirmed). Chen *et al.* [7] document non-compliance rates exceeding 60 % in WCAG 2.1 criteria for WhatsApp, Telegram, Messenger, and Instagram. The 100 % reduction on camera and photo screens validates that Phase 5

of TDDM4IoTS—defining accessibility tests before code generation—is effective for eliminating barriers in relatively simple screens. This result aligns with Ara *et al.* [47]’s recommendation that iterative testing from early development phases is the highest-impact intervention for WCAG conformance.

The messaging screen only achieved 38.1 % reduction, revealing an architectural limitation of React Native’s virtualised list pattern that generates duplicate accessibility identifiers. This limitation is not unique to the proposed system: Da Silva *et al.* [13] and Swati *et al.* [15] identified the same pattern in WhatsApp and Messenger, respectively. The problem is therefore structural to the virtualised message list paradigm and unresolved in any reported implementation.

The absence of expert manual inspection constitutes a recognised methodological limitation that may have underestimated actual barriers, particularly in the comprehensibility and robustness dimensions, whose automated evaluation is only partial even in the most advanced tools [47].

### C. Interpretation of Technological Acceptance Results

The global TAM mean ( $\bar{x}=4.20$ ;  $SD=0.95$ ) exceeds the positive acceptance threshold established in mobile accessibility TAM studies [40] (RQ2 = confirmed). The psychometric robustness of the instrument is confirmed by Cronbach’s alpha values ( $\alpha_{\text{global}}=0.866$ ;  $\alpha_{\text{PU}}=0.867$ ;  $\alpha_{\text{PEOU}}=0.898$ ), all exceeding the 0.80 threshold [41]. This supports Davis’ paradigm [39] that perceived usefulness is the dominant predictor of adoption, and is consistent with Mateos-Sanchez *et al.* [26], where AI-based conversion functionalities were the highest-rated.

The PU construct achieved the highest mean ( $\bar{x}_{\text{PU}}=4.55$ ), with PTAM002 as the highest-rated item ( $\bar{x}=4.76$ ; 84 % at level 5). This validates Elshahar *et al.* [20]’s argument that conversational speed—not conversion precision—is the critical usability factor in AAC platforms.

The PEOU construct showed greater internal dispersion ( $\bar{x}_{\text{PEOU}}=3.85$ ). PTAM007 ( $\bar{x}=3.56$ ;  $SD=1.36$ ) obtained the lowest instrument rating, with variability across the full 1–5 range. This resonates with Bleakley *et al.* [27]: commercial STT systems present barriers for users with low technology experience (40 % of the sample) related to voice-input learning curves. The “elevated expectations in populations highly dependent on accessibility tools” phenomenon, documented by Weichbroth [4], provides the most coherent interpretation of the lower perceived-ease scores.

The HJ-Biplot (Fig. 3) revealed that PTAM007 and PTAM012 vectors are orthogonal to the main cluster, capturing independent variance associated with voice interaction appropriation and complete functional mastery, respectively. This is theoretically consistent with the original TAM [39], in which ease of use moderates the effect of perceived usefulness on adoption intent, and has been documented in AAC applications by Kim *et al.* [17].

### D. Implications for Inclusive Messaging System Design

The findings allow four concrete implications for AI-driven inclusive messaging system design.

The **first implication** is that multimodal integration of TTS, STT, image captioning, and OCR in a single mobile platform is technically feasible with available open-source technologies, and produces positive technological acceptance ( $\bar{x}_{TAM}=4.20$ ;  $\bar{x}_{PU}=4.55$ ;  $\bar{x}_{PEOU}=3.85$ ) in heterogeneous sensory-disability populations, a finding without documented precedent in the literature.

The **second implication** is that asymmetric end-to-end encryption is compatible with multimodal conversion flow without degrading error rates, demonstrating for the first time in the inclusive messaging literature that privacy and accessibility are not conflicting objectives in a distributed mobile architecture. None of the reviewed AAC systems—including Talkitt [25], CapacitaBOT [26], and BellChat [21]—had reported an equivalent privacy implementation.

The **third implication** is that the voice command module via cosine similarity with cascade correction reduces, but does not eliminate, the voice navigation learning-curve barrier (PTAM007:  $\bar{x}=3.56$ ). Designing an accessible onboarding process that guides users in internalising available commands is a necessary condition for the system to achieve its full accessibility potential for users with severe visual impairment.

The **fourth implication** is that TDDM4IoTS [12] with accessibility as a verifiable acceptance criterion from Phase 5 produced a 64 % reduction in WCAG 2.1 warnings, contrasting with the total absence of accessibility evaluation in the development cycles of the reviewed CAA systems [24]–[26].

#### E. Limitations and Future Work

The present study presents four limitations.

**First, sample size:**  $n=25$  participants, though justified by clinical specificity and the projective research design [36], does not permit generalisation across the diversity of sensory disability profiles or other geographic and cultural contexts.

**Second, TTS latency:** with minimum values of 5.14 s, the system does not yet reach the 2 s conversational threshold [9].

**Third, residual messaging screen accessibility (38.1 % reduction):** architectural cause identified—duplicate identifiers in React Native virtualised lists—with a defined resolution protocol.

**Fourth, absence of sign language recognition,** which excludes deaf users whose native language is sign language. Integration of a sign language recognition module—as proposed by Saleem *et al.* [33] and Kothadiya *et al.* [35]—is the most significant functional extension.

#### F. Alignment with Sustainable Development Goals

This work aligns with two UN 2030 Agenda SDGs. Regarding **SDG 10** (Reduced Inequalities), the system demonstrates that mobile AI technology can reduce communicative asymmetries that marginalise 124,801 people with sensory disabilities in Ecuador—a figure the WHO scales to 15 % of the world population [1]—from autonomous digital communication. Regarding **SDG 9** (Industry, Innovation, and Infrastructure), the system proposes a microservices architecture deployable on low-cost hybrid infrastructure (institutional server plus cloud instance), reducing institutional adoption barriers in resource-constrained contexts such as UTEQ.

## VI. CONCLUSION

This article has presented the design, implementation, and empirical evaluation of an AI-powered inclusive mobile messaging system conceived to serve visual and hearing sensory profiles through the integrated orchestration of TTS, STT, automatic image description, and OCR modules, with asymmetric end-to-end encryption and complete voice command navigation. The findings allow four substantive conclusions and five future research directions.

The **first conclusion** concerns the technical viability of multimodal integration with guaranteed privacy (RQ1=confirmed). All four conversion modules recorded zero error rates across all test conditions, with verified transmission integrity per condition ( $< 1\%$  difference in STT;  $< 0.2\%$  in video). The coexistence of the accessible conversion flow with asymmetric end-to-end encryption—without degrading error rate—demonstrates for the first time in the inclusive messaging literature that privacy and accessibility are not conflicting objectives in a distributed mobile architecture. The minimum TTS latency of 5.14 s (SAPI5, short text) remains the primary technical limitation by exceeding the 2 s conversational threshold; its resolution via non-autoregressive TTS models or predictive voice-segment caching represents the highest-urgency improvement for future versions.

The **second conclusion** concerns positive technological acceptance in a heterogeneous disability population (RQ2=confirmed). The 12-item TAM questionnaire—organised in its classic PU and PEOU constructs [39]—applied to  $n=25$  participants produced  $\bar{x}=4.20$  ( $SD=0.95$ ), exceeding the positive acceptance threshold  $\bar{x}>4.0$  [40]. Psychometric robustness was confirmed ( $\alpha_{global}=0.866$ ;  $\alpha_{PU}=0.867$ ;  $\alpha_{PEOU}=0.898$ ; all above 0.80 [41]). PU achieved  $\bar{x}_{PU}=4.55$ , with PTAM002 ( $\bar{x}=4.76$ ; 84 % at level 5) as the highest-valued item. The voice navigation initial learning curve (PTAM007:  $\bar{x}=3.56$ ;  $SD=1.36$ ) and complete functional mastery (PTAM012:  $\bar{x}=3.48$ ) are the two priority improvement dimensions of the PEOU construct ( $\bar{x}_{PEOU}=3.85$ ).

The **third conclusion** concerns the efficacy of the TDDM4IoTS iterative cycle for systematic accessibility improvement (RQ3=confirmed). The methodology [12] with accessibility as a verifiable acceptance criterion from Phase 5 produced a global 64.0 % reduction in WCAG 2.1 warnings (75 to 27), with total elimination on camera and photo screens (100 %) and reductions  $>70\%$  on settings (77.8 %) and main screens (81.8 %). The residual limitation on the messaging screen (38.1 %) represents a traceable technical debt with a defined resolution protocol.

The **fourth conclusion** concerns the resolution of technological fragmentation documented in the literature. The AI components for accessibility—neural TTS, transformer-based STT, image captioning, and OCR—which the literature has developed in isolation [9], [10], [30]–[32] are integrable in a coherent, stable, privacy-guaranteed mobile messaging architecture on low-cost institutional hybrid infrastructure.

Five high-priority future research directions follow: (i) optimising TTS latency via non-autoregressive architectures (Fast-



Speech2, VITS) or predictive voice-segment caching to reach the 2 s conversational threshold [9]; (ii) designing an accessible onboarding flow with personalised voice command calibration to the user's disfluency profile, following the adaptive training paradigm of Costanzo *et al.* [25]; (iii) integrating sign language recognition as an additional input/output modality, taking the LSTM-GRU models of Kothadiya *et al.* [35] and the multilingual systems of Saleem *et al.* [33] as references; (iv) architectural refactoring of the messaging screen with dynamic unique accessibility identifiers per list element to eliminate the 13 residual warnings and raise global system satisfaction (PTAM012:  $\bar{x}=3.48$ ); (v) expanding the study to samples stratified by disability type, age range, and mobile connectivity conditions (4G/LTE low-bandwidth networks) to increase results generalisability and identify technology acceptance moderation factors specific to each sensory profile.

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